

Name: Sujal Suryawanshi

PRN : 202501040037

PRACTICAL: 01

1.1.1. Calculate Momentum

01:31

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula:

where:

is the mass of the object (in kilograms).

is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

```
mass = float(input())  
velocity = float(input())  
momentum = mass * velocity  
print(f"{momentum:.2f}kgm/s")
```

The screenshot displays a code editor window titled "calculate..." with a "Submit" button in the top right. The code in the editor is as follows:

```
1 mass = float(input())  
2 velocity = float(input())  
3 momentum = mass * velocity  
4 print(f"{momentum:.2f}kgm/s")
```

Below the code, the execution results are shown. The average time is 0.010 s (9.75 ms) and the maximum time is 0.012 s (12.00 ms). The test results indicate that 2 out of 2 shown test case(s) passed and 2 out of 2 hidden test case(s) passed.

Test case 1 (8 ms) is shown with the following expected and actual outputs:

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

Test case 2 (9 ms) is shown with the following expected and actual outputs:

Expected output	Actual output
2.3	2.3
4.8	4.8
11.04kgm/s	11.04kgm/s

The bottom of the window shows tabs for "Terminal" and "Test cases".

1.1.2. Conditional Calculation Based on the Number of Digits

03:48

Write a Python program that accepts an integer as input. Depending on the number of digits in .

Constraints:

$$1 \leq n \leq 999$$

Input Format:

- The input consists of a single integer .

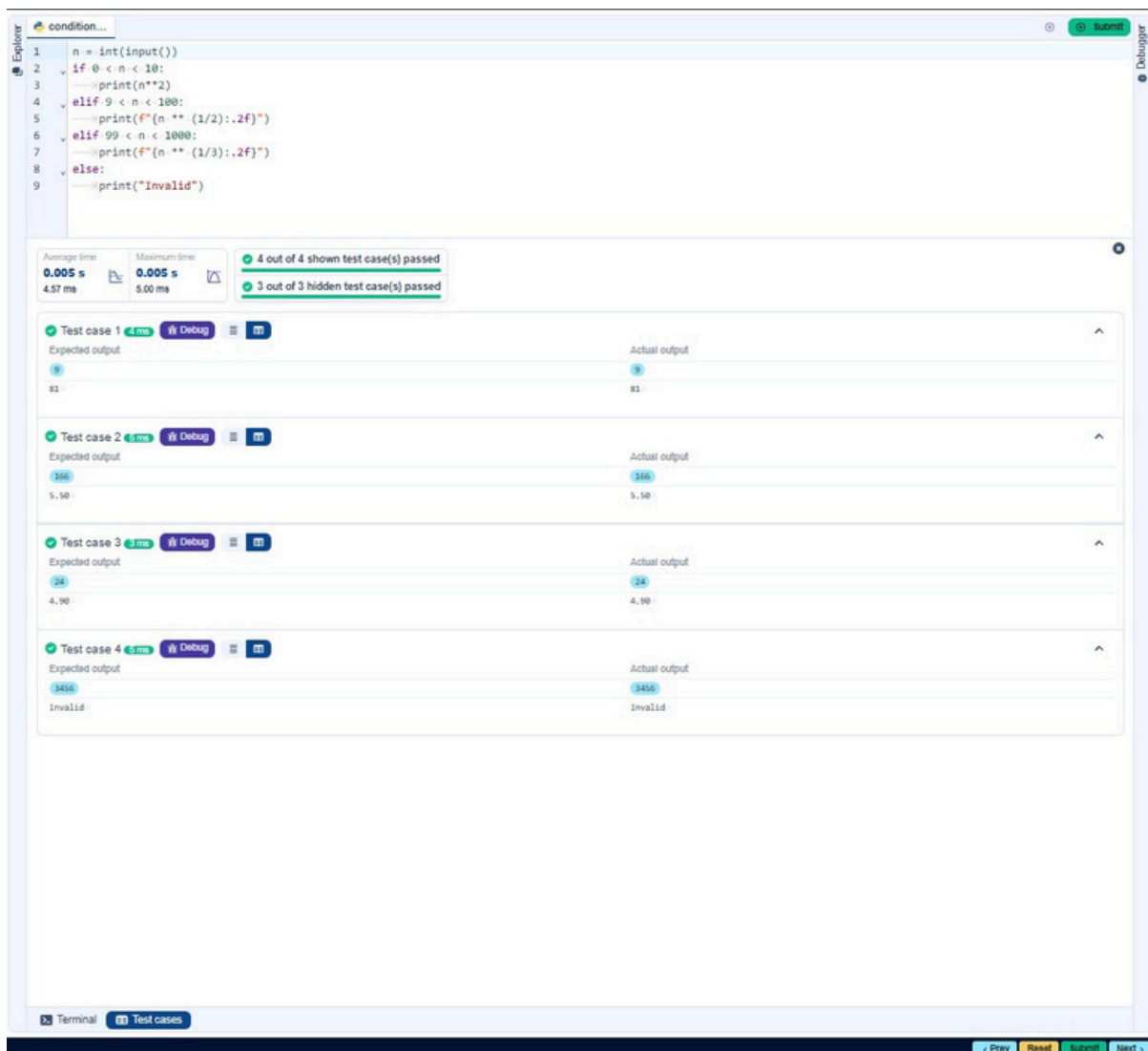
Output Format:

- If is a single-digit number, print its square.
- If is a two-digit number, print its square root (rounded to two decimal places).
- If is a three-digit number, print its cube root (rounded to two decimal places).

```
n = int(input())
```

- if $0 < n < 10$:
- `print(n**2)`
- elif $9 < n < 100$:

- `print(f"{n ** (1/2):.2f}")`
- `elif 99 < n < 1000:`
- `print(f"{n ** (1/3):.2f}")`
- `else:`
- `print("Invalid")`



Output Format:

- If is a single-digit number, print its square.
- If is a two-digit number, print its square root (rounded to two decimal places).
- If is a three-digit number, print its cube root (rounded to two decimal places).

1.1.3. Days between Two Dates

- 05:19
- Write a Python program to calculate number of days between two dates.
-
- Input Format:
- The first line contains the first date in the format .
- The second line contains the second date in the format .
-
- Output Format:
- An integer representing the number of days between the given dates.
-
- Note:
- The first date should always be considered earlier than the second date.
-

```
from datetime import date
```

- `date1_input = input().strip()`
- `date2_input = input().strip()`

- `y1, m1, d1 = map(int, date1_input.split('-'))`
- `y2, m2, d2 = map(int, date2_input.split('-'))`
- `date1 = date(y1, m1, d1)`
- `date2 = date(y2, m2, d2)`
- `difference = date2 - date1`
- `print(difference.days)`

noOfDay...

```

1 from datetime import date
2 date1_input = input().strip()
3 date2_input = input().strip()
4 y1, m1, d1 = map(int, date1_input.split('-'))
5 y2, m2, d2 = map(int, date2_input.split('-'))
6 date1 = date(y1, m1, d1)
7 date2 = date(y2, m2, d2)
8 difference = date2 - date1
9 print(difference.days)

```

Average time: 0.007 s (7.25 ms) | Maximum time: 0.010 s (10.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (10 ms) [Debug]

Expected output	Actual output
2014-07-02	2014-07-02
2014-11-02	2014-11-02
123	123

Test case 2 (6 ms) [Debug]

Expected output	Actual output
2014-07-02	2014-07-02
2014-07-11	2014-07-11
9	9

Terminal | Test cases

< Prev | Reset | Submit | Next >

```
n = int(input())
```

- if $0 < n < 10$:
- `print(n**2)`
- elif $9 < n < 100$:
- `print(f"{n ** (1/2):.2f}")`
- elif $99 < n < 1000$:
- `print(f"{n ** (1/3):.2f}")`
- else:
- `print("Invalid")`

The screenshot shows a Python IDE with a code editor and a test runner interface. The code in the editor is as follows:

```
1 n = int(input())
2 if 0 < n < 10:
3     print(n**2)
4 elif 9 < n < 100:
5     print(f"{n ** (1/2):.2f}")
6 elif 99 < n < 1000:
7     print(f"{n ** (1/3):.2f}")
8 else:
9     print("Invalid")
```

Below the code editor, the test runner shows the following statistics:

- Average time: 0.005 s
- Maximum time: 0.005 s
- 4 out of 4 shown test case(s) passed
- 3 out of 3 hidden test case(s) passed

The test runner also displays four test cases with their expected and actual outputs:

Test Case	Expected output	Actual output
Test case 1	9	9
Test case 2	166	166
Test case 3	24	24
Test case 4	Invalid	Invalid

1.1.4. Reverse a Number

01:16

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.


```
num = int(input())
reversed_num = 0
while num > 0:
    digit = num % 10
    reversed_num = reversed_num * 10 + digit
    num = num // 10
```

```
print(reversed_num)
```

The screenshot displays a Python code editor with a file named 'reverseN...'. The code is as follows:

```
1 num = int(input())
2 reversed_num = 0
3 while num > 0:
4     digit = num % 10
5     reversed_num = reversed_num * 10 + digit
6     num = num // 10
7 print(reversed_num)
```

Below the code, the execution results are shown:

- Average time: 0.003 s (3.20 ms)
- Maximum time: 0.004 s (4.00 ms)
- 2 out of 2 shown test case(s) passed
- 3 out of 3 hidden test case(s) passed

Two test cases are detailed:

- Test case 1** (4 ms):
 - Expected output: 5367, 7635
 - Actual output: 5367, 7635
- Test case 2** (2 ms):
 - Expected output: 0132, 231
 - Actual output: 0132, 231

At the bottom, there are buttons for 'Terminal', 'Test cases', and navigation controls: '< Prev', 'Reset', 'Submit', and 'Next >'.

1.1.5. Multiplication Table

01:38

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

```
n=int(input())
```

```
for i in range(1,11):
```

```
print(f"{n} x {i} = {n * i}")
```

The screenshot shows a Python IDE with a file named 'multiplica...'. The code is as follows:

```
1 n=int(input())
2 for i in range(1,11):
3     print(f"{n} x {i} = {n * i}")
```

Below the code editor, the execution results are displayed. The average time is 0.004 s and the maximum time is 0.006 s. The program passed 2 out of 2 shown test case(s) and 2 out of 2 hidden test case(s).

Test case 1 (4 ms) shows the expected output and actual output for n=8. The output is a multiplication table for 8:

Expected output	Actual output
8 x 1 = 8	8 x 1 = 8
8 x 2 = 16	8 x 2 = 16
8 x 3 = 24	8 x 3 = 24
8 x 4 = 32	8 x 4 = 32
8 x 5 = 40	8 x 5 = 40
8 x 6 = 48	8 x 6 = 48
8 x 7 = 56	8 x 7 = 56
8 x 8 = 64	8 x 8 = 64
8 x 9 = 72	8 x 9 = 72
8 x 10 = 80	8 x 10 = 80

Test case 2 (4 ms) shows the expected output and actual output for n=5. The output is a multiplication table for 5:

Expected output	Actual output
5 x 1 = 5	5 x 1 = 5
5 x 2 = 10	5 x 2 = 10
5 x 3 = 15	5 x 3 = 15
5 x 4 = 20	5 x 4 = 20
5 x 5 = 25	5 x 5 = 25
5 x 6 = 30	5 x 6 = 30
5 x 7 = 35	5 x 7 = 35
5 x 8 = 40	5 x 8 = 40
5 x 9 = 45	5 x 9 = 45
5 x 10 = 50	5 x 10 = 50

1.2.1. Pass or Fail

01:11

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer , the number of courses.
- The second input will be integers representing the marks of the student in each of the courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Input number of courses

```
n = int(input())
```

Input marks

```
marks = list(map(int, input().split()))
```

Check if any subject is failed

if any(mark < 40 for mark in marks):

```
    print("Fail")
```

else:

Calculate aggregate percentage

```
aggregate = sum(marks) / n
```

```
#Print aggregate percentage formatted to two decimal places
```

```
print(f"Aggregate Percentage: {aggregate:.2f}")
```

```
#Determine grade
```

```
if aggregate > 75:
```

```
    grade = "Distinction"
```

```
elif 60 <= aggregate < 75:
```

```
    grade = "First Division"
```

```
elif 50 <= aggregate < 60:
```

```
    grade = "Second Division"
```

```
elif 40 <= aggregate < 50:
```

```
    grade = "Third Division"
```

```
print(f"Grade: {grade}")
```

The screenshot displays a code execution interface with the following details:

- Performance Metrics:**
 - Average time: 0.006 s (6.20 ms)
 - Maximum time: 0.017 s (17.00 ms)
- Test Results Summary:**
 - 5 out of 5 shown test case(s) passed
 - 5 out of 5 hidden test case(s) passed
- Test Case 1 (17 ms):**
 - Expected output:** 4, 55 65 78 89, Aggregate Percentage: 71.75, Grade: First Division
 - Actual output:** 4, 55 65 78 89, Aggregate Percentage: 71.75, Grade: First Division
- Test Case 2 (7 ms):**
 - Expected output:** 5, 56 78 97 86 93, Aggregate Percentage: 82.00, Grade: Distinction
 - Actual output:** 5, 56 78 97 86 93, Aggregate Percentage: 82.00, Grade: Distinction

1.2.2. Fibonacci Series

32:17

Write a Python program that uses recursion to print the first terms of the Fibonacci series.

Input Format:

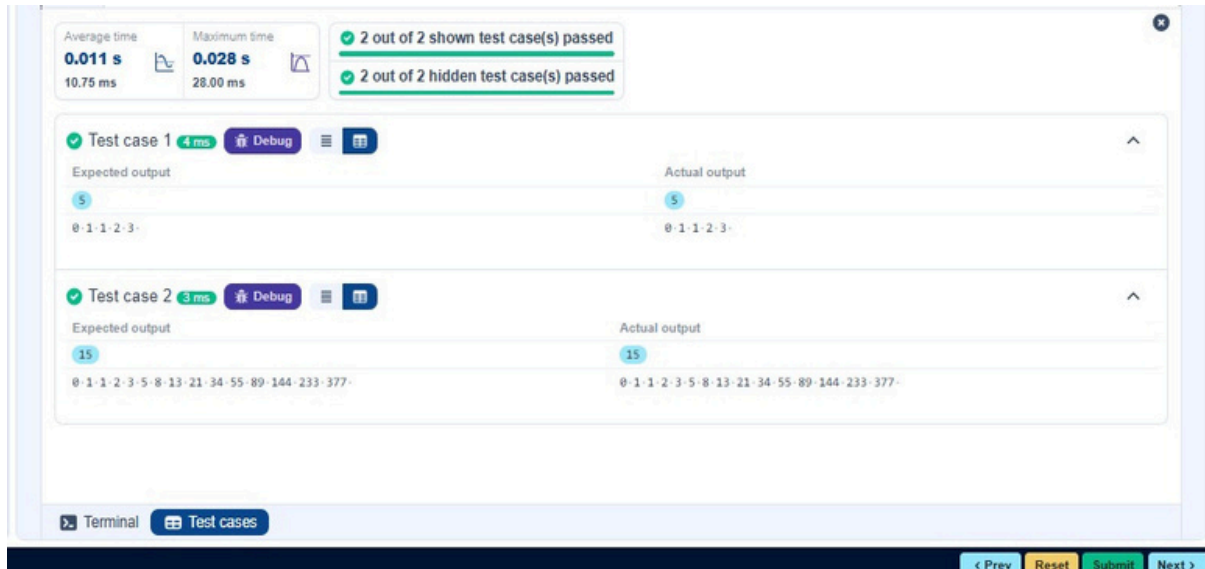
- A single integer representing the number of terms to generate.

Output Format:

- A single line containing the first terms of the Fibonacci sequence, separated by spaces.

```
def fibonacci(n):  
    if n==1:  
        return 0  
    elif n==2:  
        return 1  
    else:  
        return fibonacci(n-1) + fibonacci(n- 2)
```

```
n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
```



1.2.3. Pattern - 1

00:50

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.


```
n = int(input())
for i in range(1, n + 1):
    for j in range(i):
        print("*", end=" ")
```

print()

```
1 n = int(input())
2 for i in range(1, n + 1):
3     for j in range(i):
4         print("*", end=" ")
5     print()
```

Average time: 0.006 s (6.00 ms) | Maximum time: 0.008 s (8.00 ms)

2 out of 2 shown test case(s) passed
4 out of 4 hidden test case(s) passed

Test case 1 (7 ms) [Debug] [Test cases]

Expected output	Actual output
5	5
* , *	* , *
* , * , *	* , * , *
* , * , * , *	* , * , * , *
* , * , * , * , *	* , * , * , * , *

Test case 2 (6 ms) [Debug] [Test cases]

Expected output	Actual output
4	4
* , *	* , *
* , * , *	* , * , *
* , * , * , *	* , * , * , *

Terminal | Test cases

< Prev | Reset | Submit | Next >

1.2.4. Pattern - 2

01:10

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

```
n = int(input())
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=" ")
```

print()

numberP...

Submit

Explorer

Debugger

```
1 n = int(input())
2 for i in range(1, n + 1):
3     for j in range(1, i + 1):
4         print(j, end=" ")
5     print()
```

Average time
0.004 s
3.75 ms

Maximum time
0.005 s
5.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output

Actual output

5	5
1	1
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5

Test case 2 3 ms Debug

Expected output

Actual output

8	8
1	1
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5
1 2 3 4 5 6	1 2 3 4 5 6
1 2 3 4 5 6 7	1 2 3 4 5 6 7
1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8

Terminal

Test cases

< Prev

Reset

Submit

Next >